9 November 2026

Dear Editors,

Please consider our manuscript, “Tidal potential measured with distant optical clocks”, for publication as an Article in *Nature*.

Optical clocks can measure changes in gravitational potential through their relative ticking rates. Using this capability to study Earth requires separating overlapping geophysical contributions and establishing how clock noise and intermittent operation limit their measurement. Astronomical tides provide a predictable signal for addressing these questions, while their proximity to daily laboratory variations makes attribution challenging.

We compare a strontium clock in coastal Shanghai with an ytterbium clock in inland Wuhan across a 692-km baseline, connected by a 1,350-km fibre link. Five campaigns spanning 353 days yield 545.9 hours of observations in the main analysis. The measured response to the combined solid-Earth and ocean-loading prediction is 0.96 ± 0.11, and the lunar semidiurnal group response is 1.00 ± 0.13. The amplitude spectrum resolves peaks near the principal lunar and solar semidiurnal frequencies, with equivalent-height amplitudes of 4.87 and 2.87 cm. Conditional fits and sampling-response tests show that each contribution remains identifiable when the other is controlled. Additional analyses quantify sensitivity to specified daily disturbances and ocean-loading assumptions.

We develop a signal-to-noise model that combines signal strength, clock noise and actual sampling. The model accounts for the lunar precision and helps explain why the solar-frequency, diurnal and separate ocean-loading contributions are less precisely constrained. It relates component precision to observing duration, phase coverage and parameter coupling. Its exposure scaling is conditional on the noise and sampling efficiency of future observations.

The work will interest Nature’s readers in quantum metrology, gravitation and geophysics. It demonstrates how distant optical clocks can distinguish components of a time-varying geophysical signal and quantify the precision available for each. Characterizing this tidal background has potential applications to optical-clock networks supporting the redefinition of the second and the unification of height systems.

The 274-hour summer record is shared with the absolute frequency-ratio study by Hou and colleagues (arXiv:2610.34250). This relationship is disclosed in the manuscript; the present work uses the extended record for tidal-component analysis. We also discuss the related European-network observations by Stahl and colleagues (arXiv:2610.39635) and explain the complementary focus on separating lunar and solar semidiurnal contributions along a coastal–inland baseline.

[Author confirmation required: publication status, exclusive consideration, approval of the manuscript and submission by all authors, and any related submissions or prior editorial discussions.]

Thank you for considering our manuscript.

Yours sincerely,

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